NAME: ____ CLASS: _________ INDEX: __________

CATHOLIC JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATIONS
Higher 1

BIOLOGY
8875/01

Paper 1 Multiple Choice

1 hour

Additional Materials: Multiple Choice Answer Sheet

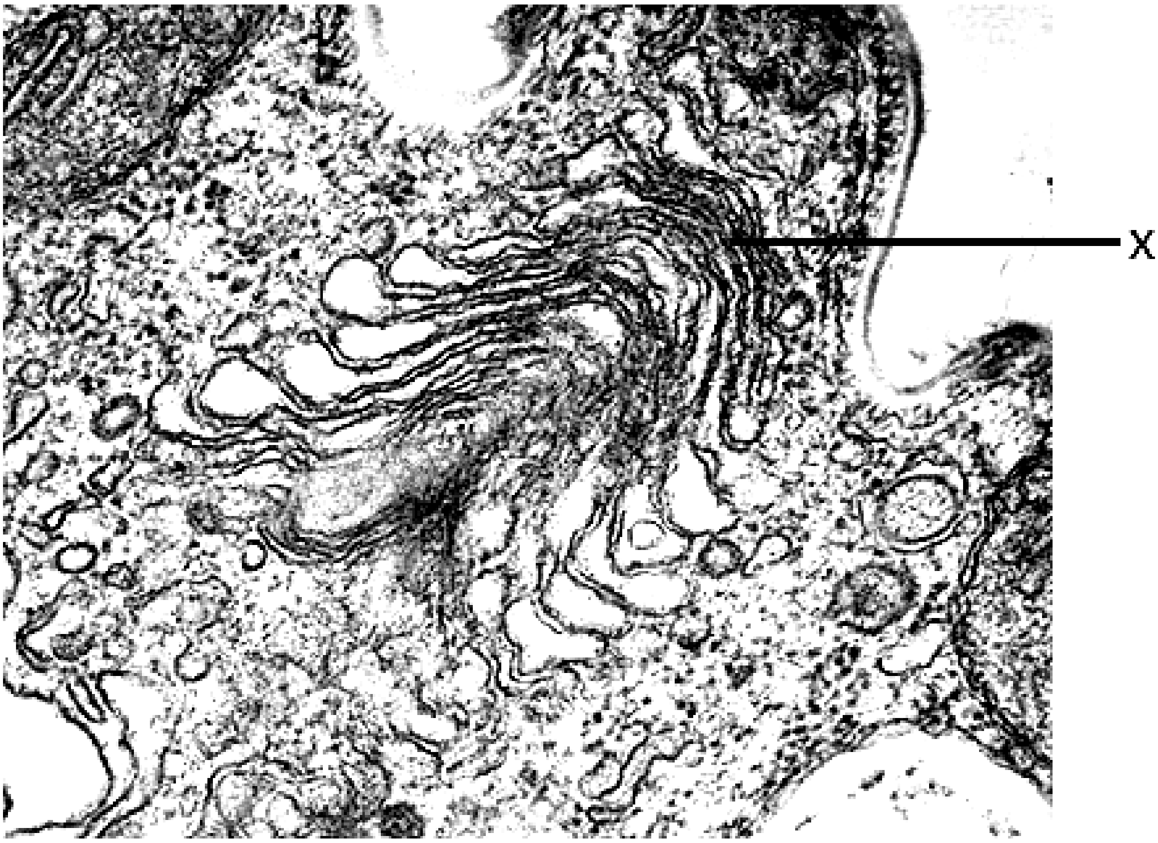
READ THESE INSTRUCTIONS FIRST
Write in soft pencil.
Do not use staples, paper clips, highlighters, glue or correction fluid.
Write and/or shade your name, NRIC / FIN number and HT group on the Answer Sheet in the spaces provided unless this has been done for you.

There are **30 MCQs** in this paper. Answer **all** questions. For each question, there are four possible answers, **A, B, C** and **D**.
Choose the **one** you consider correct and record your choice in **2B pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.
Calculators may be used.

1. Which of the following are the most likely consequences for a cell lacking Structure **X**?



1. The cell dies because it is unable to make glycoproteins to detect stimuli from its environment.

2. The cell dies from a lack of enzymes to digest food taken in by endocytosis.

3. The cell dies from the accumulation of worn out organelles within itself.

4. The cell is unable to reproduce itself.

5. The cell is unable to export its enzymes or peptide hormones.

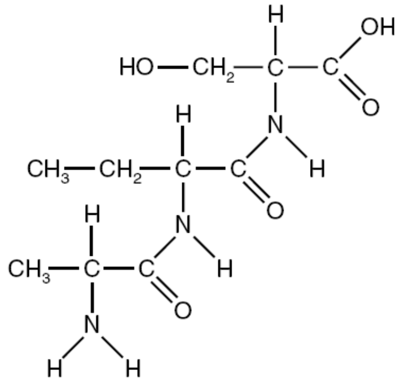
A **2** and **3**

B **2** and **5**

C **3**, **4** and **5**

D **1**, **2**, **3** and **5**

2. The diagram below shows a biological molecule found in the living cells.



The following are some statements concerning the molecule shown above.

1 It is soluble in water.
2 Double bonds between the carbon and oxygen atoms increase membrane fluidity.
3 It can act as a buffer against extreme pH change.
4 It can undergo condensation to form a polymer.

Which of the above statements are true about this molecule?

A **1** and **3**

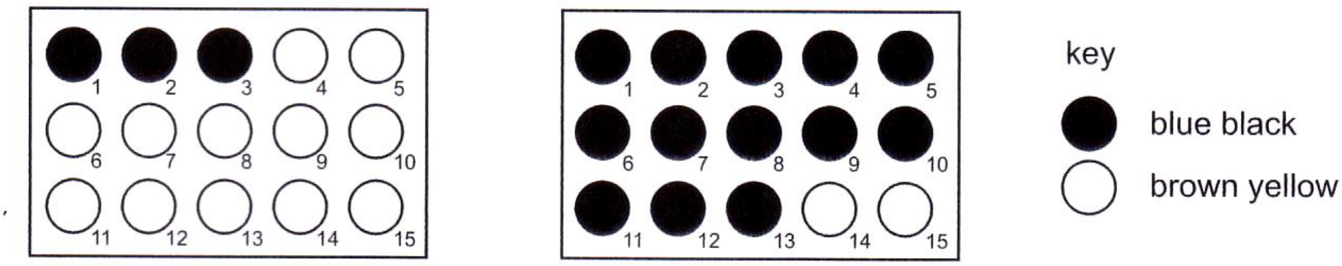
B **1** and **4**

C **2** and **3**

D **2** and **4**

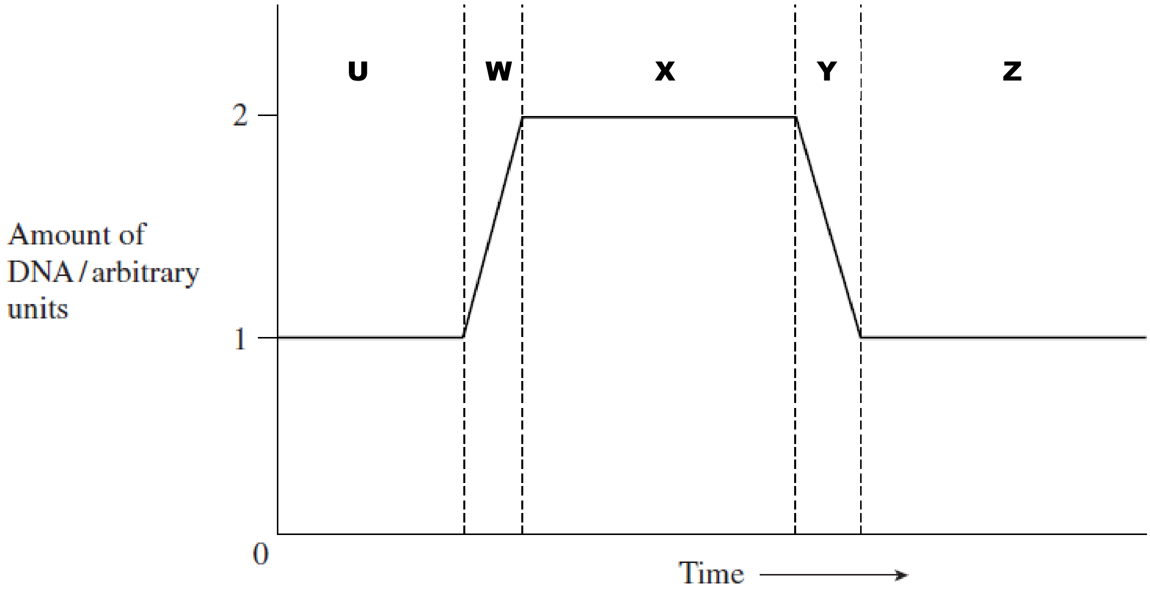
3. An experiment was carried out to investigate the digestion of starch using amylase at two different temperatures. A sample was removed from each mixture at 15 second intervals and place onto a spotting tile well containing two drops of iodine in KI solution. The results are shown in the diagram.

Which shows the correct temperatures and times for the complete digestion of starch?



	temperature / °C	**time / s**
	30	3.15
A	10	0.45
	30	45
B	10	195
	10	45
C	30	195
D	10	3.15
	30	0.45

4. The graph shows changes in the amount of DNA in a cell during one cell cycle. The letters U – Z marks out the different phases in the cell cycle.



Many drugs that are used to treat cancer work at different time periods during the cell cycle.

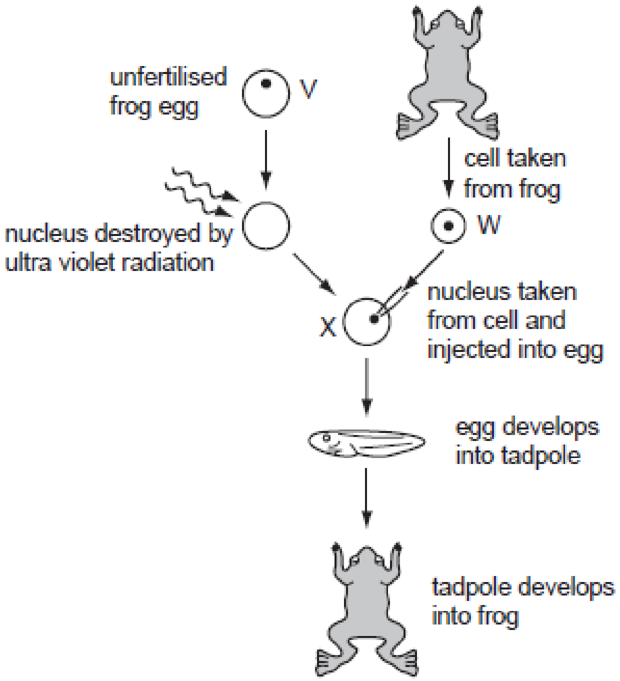
(i) Cisplatin binds to DNA and stops free DNA nucleotides from joining together.

(ii) Drug B stops spindle fibres from shortening.

With reference to the cell cycle above, determine where these 2 drugs work.

	Cisplatin	**Drug B**
A	**W**	**X**
B	**W**	**Y**
C	**U**	**X**
D	**U**	**Z**

5. The diagram shows how genetically identical frogs can be developed from unfertilised frog eggs. The diploid number for frogs is 26.



Which combination of numbers correctly identifies the number of chromosomes in each type of cell?

	V	W	X
A	13	13	26
B	13	26	13
C	13	26	26
D	26	26	13

6. The table below shows a list of characteristics displayed by mutant strains of *E.coli* during DNA replication and the possible reasons.

No	Characteristics	Enzymes or functions affected by mutation
1	Okazaki fragments accumulate and DNA synthesis is never completed	DNA ligase activity is missing
2	Supercoils are found to remain at the flanks of the replication bubbles	DNA helicase is hyperactive
3	Synthesis is very slow.	DNA polymerase keeps dissociating from the DNA and has to re-associate
4	No initiation of replication occurs.	A-T rich region at origin of replication deleted

Which of the reasons correctly explain the characteristics displayed by the mutant *E. coli* strains?

A 2 and 3

B 1 and 3

C 1, 3, and 4

D All the above

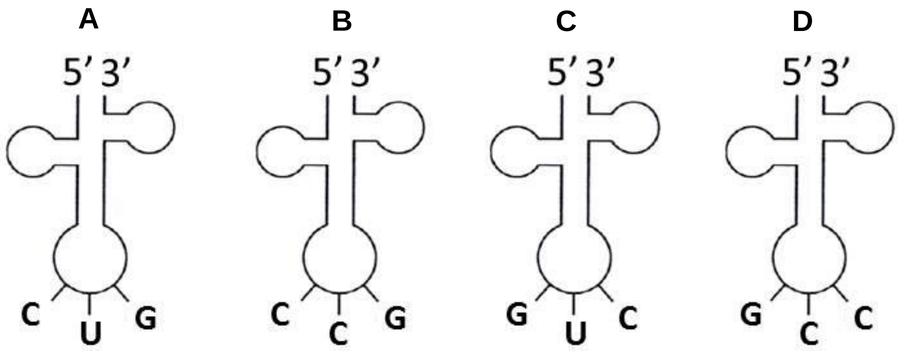
7. Part of the amino acid sequence of protein X in normal and disease patients are shown below.

Protein X in normal patient	**Protein X in disease patient**
…Lys-His-Pro-Asp-Thr…	…Lys-His-Pro-Gly-Thr…

mRNA codons for the amino acids shown in the sequence are as follows:

His: CAU, CAC	Asp: GAU, GAC	Lys: AAA, AAG
Thr: ACU, ACC, ACA, ACG	Gly: GGU, GGC, GGA, GGG	Pro: CCU, CCC, CCA, CCG

Which transfer RNA is involved in forming the modified part of protein X?



8. Listed below are some antibacterial drugs that can affect the synthesis of bacterial proteins.

Anti-microbial drug	Rifampicin	Streptomycin	Tetracycline
Mode of action	Binds to RNA polymerase	mRNA misread during translation	Prevents binding of tRNA to ribosome

Which row shows the correct effects of these drugs on bacterial protein synthesis?

	Rifampicin	**Streptomycin**	**Tetracycline**
A	Defective protein synthesised	mRNA does not bind to ribosome	Polymerization of amino acids does not occur
B	mRNA not synthesised	Defective protein synthesised	Polymerization of amino acids does not occur
C	mRNA not synthesised	mRNA does not bind to ribosome	Transcription prevented
D	Transcription prevented	Defective protein synthesised	mRNA does not bind to ribosome

9. What do chromosomal aberrations and gene mutations have in common and how are they different?

	Similarity	Difference
A	Both may involve addition of nucleotides.	Gene mutations always produce dominant alleles but not chromosomal aberrations.
B	Both may not result in disorders.	Gene mutations do not involve inversions but inversions of segments of chromosomes do occur.
C	Both affect nucleotide sequence in DNA molecule.	Gene mutations occur within a chromosome but chromosomal aberrations may occur across chromosomes.
D	Both may not result in a difference in protein expression.	Chromosomal aberrations are more harmful than gene mutations.

10. In poultry, males have two X chromosomes and females have one X chromosome and one Y chromosome. The gene for feather-barring is sex-linked. The allele for barred feathers is dominant to the allele for non-barred feathers. A non-barred male is crossed with a barred female.

What ratio of offspring would be expected?

A 1 barred male : 1 barred female

B 1 non-barred male : 1 non-barred female

C 1 barred male : 1 non-barred female

D 1 non-barred male : 1 barred female

11. In cattle, the gene responsible for normal development of hair and teeth, ectodysplasin 1 (*ED1*) is located on the X chromosome. Mutations in the *ED1* gene result in a rare genetic disorder, anhidrotic ectodermal dysplasia. Another character, the presence of horns, is determined by a gene on an autosome. The allele for the absence of horns (**H**) is dominant and the allele for the presence of horns (**h**) is recessive.

A horned bull with anhidrotic ectodermal dysplasia was mated on several occasions to the same female. A large number of offspring consisting of males and females in equal numbers in all combinations of phenotypes are shown in the table.

Offspring phenotypes
No anhidrotic ectodermal dysplasia, horns present
No anhidrotic ectodermal dysplasia, horns absent
Anhidrotic ectodermal dysplasia, horns present
Anhidrotic ectodermal dysplasia, horns absent

If $\mathbf{X^E}$ represents an X chromosome carrying the normal *ED1* allele and $\mathbf{X^e}$ represents an X chromosome carrying the *ED1* allele for anhidrotic ectodermal dysplasia, what is the genotype of the female parent?

A $\mathbf{X^EX^EHH}$

B $\mathbf{X^EX^EHh}$

C $\mathbf{X^EX^eHH}$

D $\mathbf{X^EX^eHh}$

12. The table below shows the blood group phenotypes resulting from crosses between different genotypes.

Genotype of 1st parent	Genotype of 2nd parent	
	I^AI^A	I^AI^O
I^AI^O	A	A and O
I^BI^O	A and AB	A, B, B and AB

Two parents have a son who has blood group O and phenylketonuria. One parent has blood group O and the other has blood group A. Neither parent has phenylketonuria.

What is the probability that the second child of these parents will be a girl with blood group A who does not have phenylketonuria?

A 1 in 16

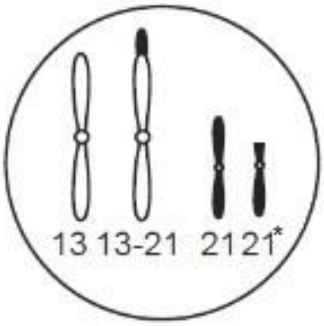
B 1 in 8

C 3 in 16

D 3 in 8

13. Down's syndrome is caused either by trisomy 21 or by translocation of a portion of chromosome 21 onto chromosome 13, forming a single chromosome 13-21. The remainder of chromosome 21 is denoted as 21*.

The diagram shows chromosomes 13 and 21 in the nucleus of a diploid (2n) cell from a male with a 13-21 translocation. This cell has a chromosome number of 46.

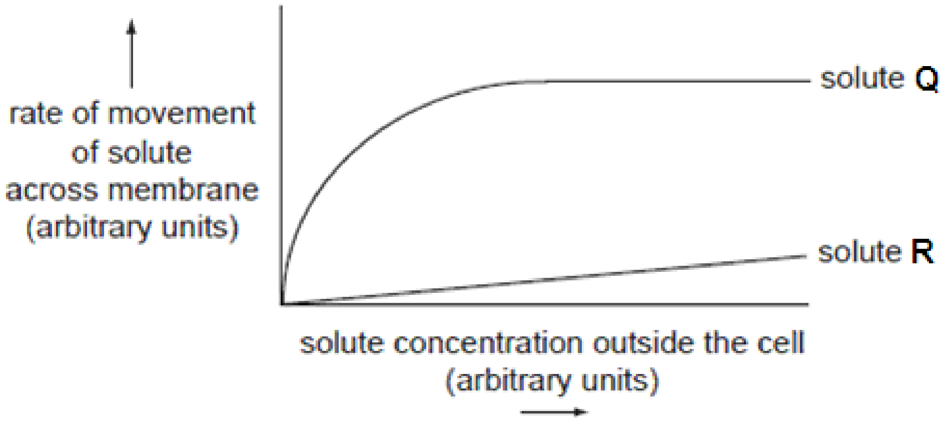


In order to produce a Down's syndrome child, which of the following is a likely outcome of fertilisation of a normal oocyte by a sperm from this male?

	Chromosomes in the sperm	Number of chromosomes
A	13 and 21*	47
B	13 – 21	46
C	13 – 21 and 21	46
D	13 – 21 and 21*	47

14. Solute **Q** moves across the plasma membrane via protein channels. Solute **R** moves across the plasma membrane by simple diffusion. The rate of movement of each solute into a cell is recorded and graphed.

The results are shown in the following graph.



It would be reasonable to conclude that

A solute **Q** is lipid soluble.

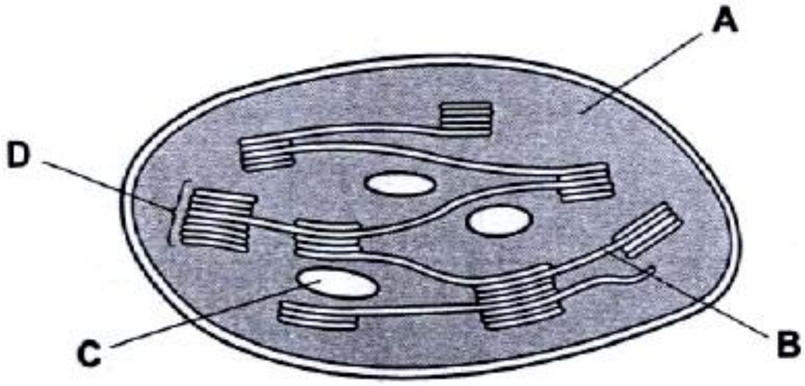
B solute **Q** saturates the protein channels.

C the gradient of the graph for solute **R** will increase as the temperature decreases.

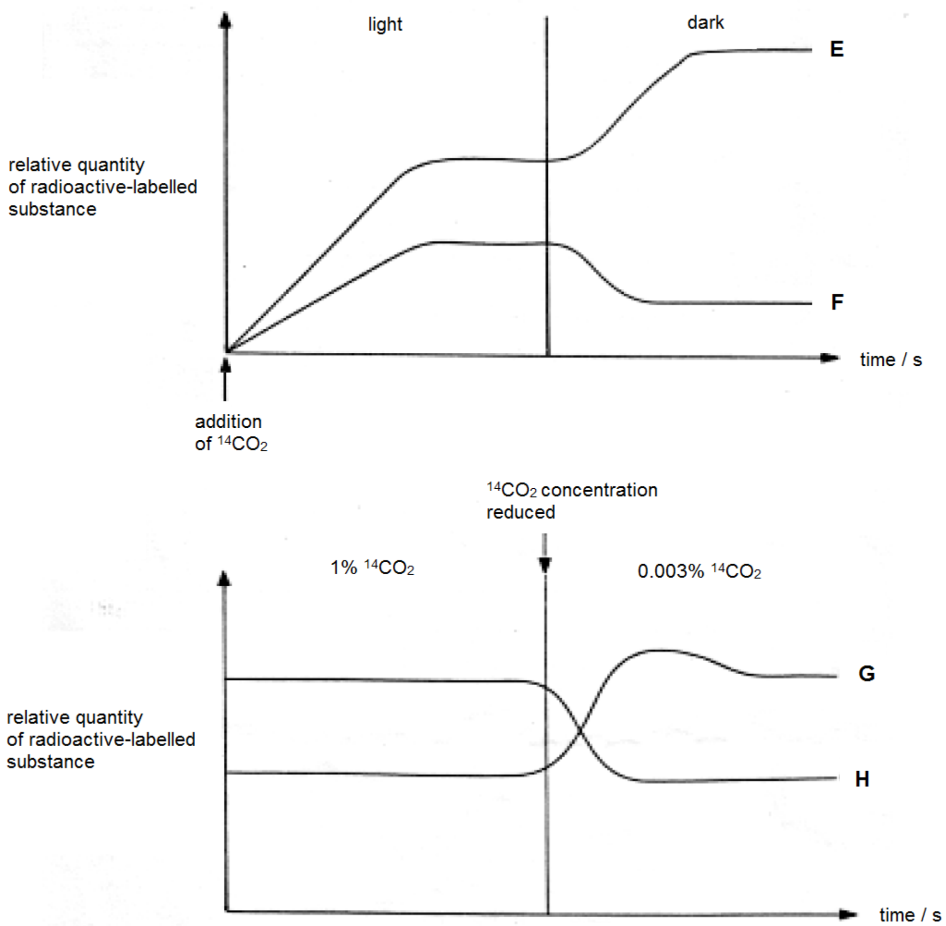
D as solute **R** is metabolised within a cell, its rate of movement across the membrane decreases.

15. The diagram shows a section through a chloroplast.

Where would the products of photophosphorylation be used?



16. The following graphs shows two separate experiments on carbon dioxide fixation in photosynthesis using a unicellular green alga and radioactive isotope ^{14}C to label the carbon dioxide. The algae were actively photosynthesizing before the start of both experiments.



Which of the following correctly identify the graphs **E**, **F**, **G** and **H**?

	E	**F**	**G**	**H**
A	RuBP	GP	GP	RuBP
B	GP	RuBP	GP	RuBP
C	GP	RuBP	RuBP	GP
D	RuBP	GP	RuBP	GP

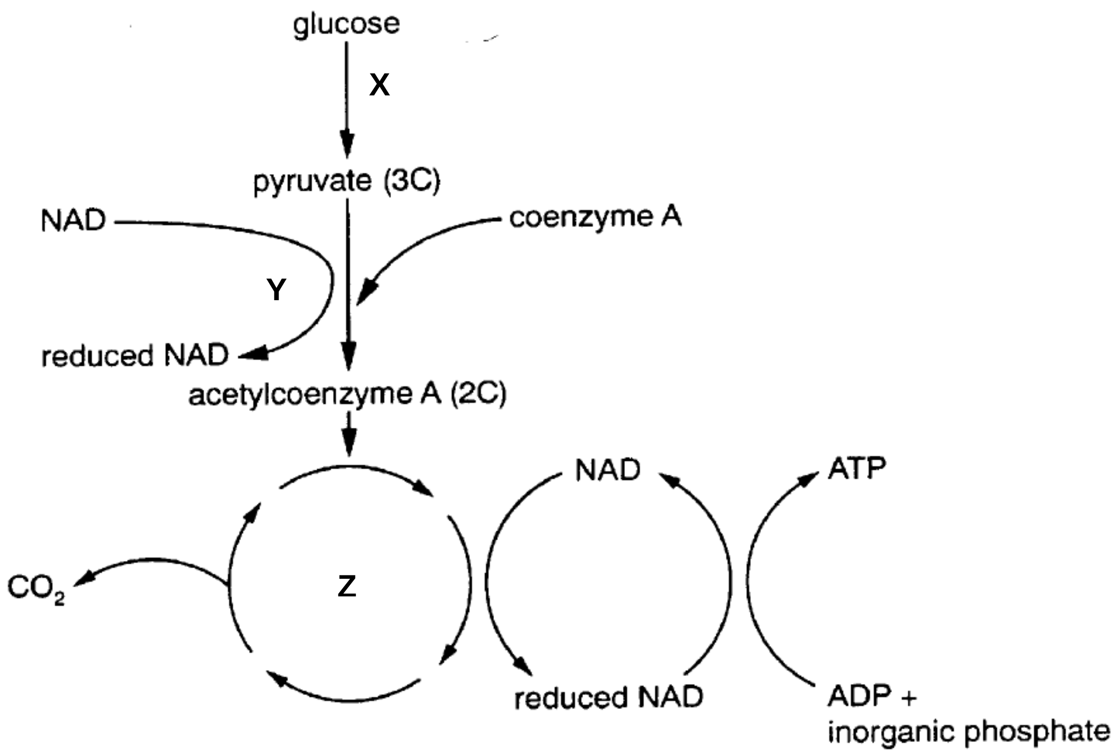
17. Rotene and oligomycin are two metabolic poisons which affect cellular respiration. The effects of rotene and oligomycin on aerobic respiration are summarised in the table.

	Ability to use glucose	**Ability to use oxygen**	**ATP yield**
Rotene	Yes	No	Decreases
Oligomycin	Yes	Yes	Decreases

Which of the following correctly identifies the specific functions of these two metabolic poisons?

	Rotene	**Oligomycin**
A	Electron transport inhibitor	Inhibits ATP synthase
B	Inhibits ATP synthase	Electron transport inhibitor
C	Dissipate proton gradient	Inhibits ATP synthase
D	Inhibits ATP synthase	Dissipate proton gradient

18. The flow chart shows a series of reactions occurring in an animal cell.



Which of the following statements correctly describes the flow chart?

A Reaction **X**, which occurs in the cytosol, is an anabolic reaction.

B Reaction **Y** involves the process of substrate-level phosphorylation, whereby pyruvate is first converted to a compound called acetyl coenzyme A.

C Reaction **Y** occurs in the cytoplasm whereas reaction **Z** occurs in the mitochondria.

D Reaction **Z** is a catabolic pathway which occurs twice for every glucose molecule to be completely oxidised.

19. Which sequence of events correctly describes evolution?

1 Differential reproduction of the spiders occurs.
2 A new selection pressure occurs.
3 Allele frequencies within the spider population change.
4 Poorly adapted spiders have decreased survivorship.

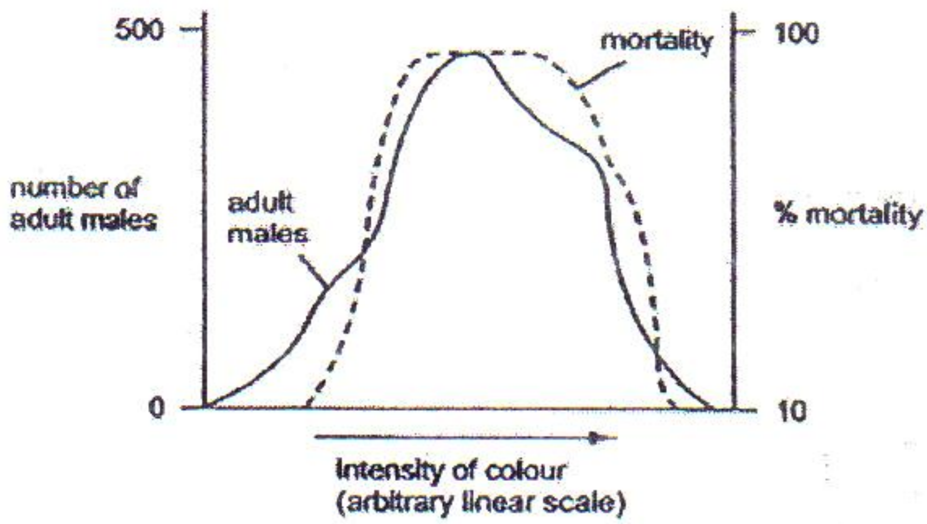
A **2 4 1 3**

B **2 4 3 1**

C **4 1 3 2**

D **4 3 1 2**

20. The graph below shows data on a population of a species of moth which shows considerable variation in colour intensity



Which conclusion can be made from this graph?

A Colour variation is environmentally induced.

B Colour variation is genetically determined.

C Extreme forms are favoured by natural selection.

D The species shows discontinuous variation with respect to colour.

21. Darwin's view of the process of evolution to form new species (speciation) has been reinforced by more recent discoveries in genetics and cell biology. This is termed the neo-Darwinian view of evolution.

In this view, which sequence of events is considered most likely to lead to speciation?

A	adaptation of population	→	competition and predation leading to natural selection	→	behavioural isolation	→	sympatric speciation
B	adaptation of population	→	competition and predation leading to natural selection	→	behavioural isolation	→	allopatric speciation
C	competition and predation leading to natural selection	→	geographical isolation	→	adaptation of isolated populations	→	sympatric speciation
D	competition and predation leading to natural selection	→	geographical isolation	→	adaptation of isolated populations	→	allopatric speciation

22. The table shows amino acid sequences from a segment of a viral protein that is found in a number of different viruses. These viruses were isolated from patients suffering from either the common cold caused by rhinovirus or flu caused by influenza. Each letter represents a single amino acid.

Virus	**Amino acid sequence**																	
Rhinovirus **1**	P	E	F	P	Y	N	A	T	T	E	P	T	K	A	V	P	F	Q
Rhinovirus **2**	P	E	F	S	Y	S	A	V	D	D	P	I	G	E	E	P	F	K
Rhinovirus **3**	P	E	F	S	Y	S	A	G	D	D	P	A	G	E	E	P	F	N
Influenza **A**	P	E	Y	H	T	P	V	M	E	P	L	D	P	F	T	M	D	K
Influenza **B**	P	E	Y	H	T	P	K	M	E	P	L	D	A	F	A	M	D	N
Influenza **C**	P	E	Y	H	T	P	V	M	E	P	L	D	V	F	D	M	D	K

Which of the following statements is consistent with the data given?

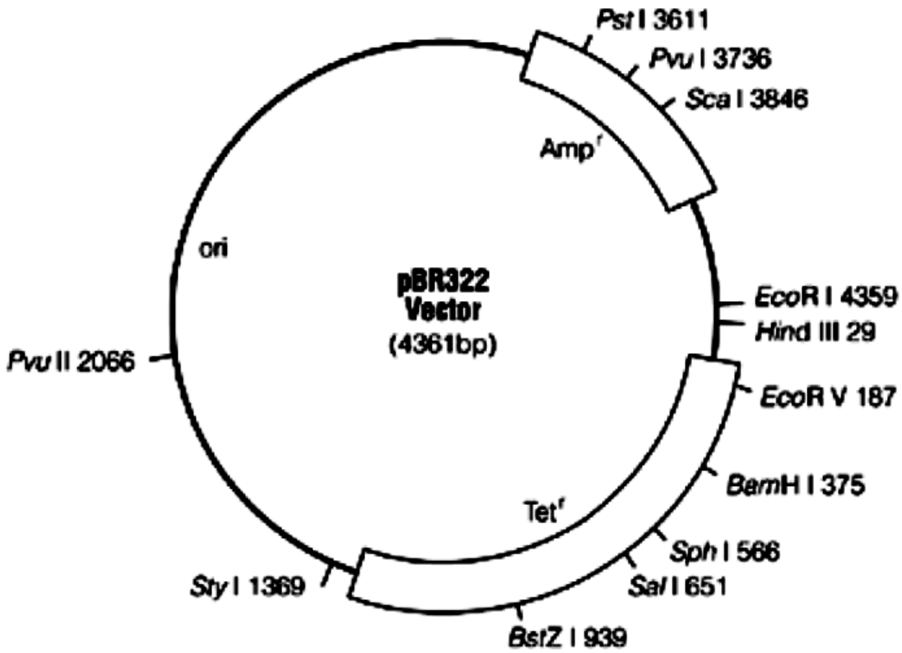
A The rhinoviruses are more closely related to each other than the influenza viruses are to each other.

B Influenza virus **A** is more closely related to influenza virus **B** than influenza virus **B** is to influenza virus **C**.

C The influenza viruses are less closely related to each other than they are to the rhinoviruses.

D Rhinovirus **2** is more closely related to rhinovirus **3** than it is to rhinovirus **1**.

23. The pBR322 vector is used to clone a eukaryotic gene, which has been digested by the restriction endonuclease *BamHI*.



Following transformation, bacterial cells were grown in four different media, as shown below:

1 Nutrient broth containing ampicillin
2 Nutrient broth containing tetracycline
3 Nutrient broth containing ampicillin and tetracycline
4 Nutrient broth without ampicillin and tetracycline

Which of the following media would bacterial cells containing the recombinant plasmids grow in?

A **4**

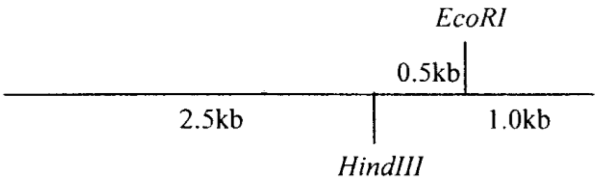
B **1** and **2**

C **2** and **3**

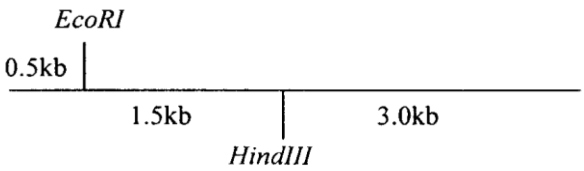
D **1** and **4**

24. Digestion of a 4 kb DNA molecule with *EcoR*I yields two fragments of 1 kb and 3 kb each. Digestion of the same molecule with *Hind*III yields fragments of 1.5 kb and 2.5 kb. Finally, digestion with *EcoR*I and *Hind*III in combination yields fragments of 0.5 kb, 1 kb and 2.5 kb. How would a restriction map indicating the positions of the *EcoR*I and *Hind*III cleavage sites look like?

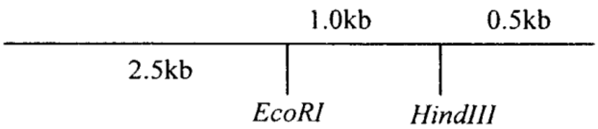
A



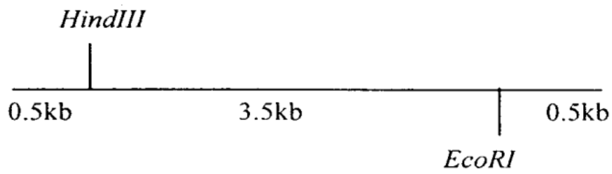
B



C



D



25. The dashed lines in the template sequence represent a long sequence of bases to be amplified.

Template

5' ATTCGGACTTG - - - - - - - - - GTCCAGCTAGAGG 3'
3' TAAGCCTGAAC - - - - - - - - - CAGGTCGATCTCC 5'

Which of the following sets of primers can be used in the PCR for the amplification of the following DNA sequence?

A **5’** GTCCAGC **3’** & **5’** CCTGAAC **3’**

B **5’** ATTCGGA **3’** & **5’** CCTCTAG **3’**

C **5’** GGACTTG **3’** & **5’** GCTGGAC **3’**

D **5’** AUUCGGA **3’** & **5’** GAUCUCC **3’**

26. Which of the following statements about the human genome project (HGP) is **false**?

A HGP aims to identify all the genes in human and to determine the DNA sequences of these genes.

B HGP aims to allow genetic testing to take place for earlier detection of genetic diseases

C HGP allows defective genes to be replaced through gene therapy

D HGP allows comparative studies to be made between humans and other organisms to identify similar genes associated with diseases.

27. What is the role of stem cells with regard to the function of adult tissues and organs?

A Stem cells are undifferentiated cells that divide asymmetrically, giving rise to one daughter cell that remains a stem cell and one daughter that will differentiate to replace damaged and worn out cells in the adult tissue or organ.

B Stem cells are embryonic cells that persist in the adult, and can give rise to all of the cell types in the body.

C Stem cells are differentiated cells that have yet to express the genes and produce proteins characteristic of their differentiated state, and do so when needed for repair of tissues and organs.

D Stem cells are fully differentiated cells that reside under the surface of epithelial tissue, in position to take over the function of the tissue when the overlying cells become damaged or worn out.

28. Which of the following regarding embryonic stem cells and hematopoietic stem cells is true?

A As embryonic stem cells develop, they turned into hematopoietic stem cells as they lose their ability to differentiate into all types of cells.

B Embryonic stem cells have more genes than hematopoietic stem cells and thus are able to form more types of cells.

C Under normal conditions, embryonic stem cells express more of their genes compared to the hematopoietic stem cells.

D Both stem cells are derived from the zygotic stem cells with the hematopoietic stem cells having a lowered telomerase activity compared to the embryonic stem cells.

29. Some scientists are concerned about the release of genetically modified microorganisms into their natural habitat.

What is the most likely reason for this concern?

A The microorganisms may reproduce quickly.

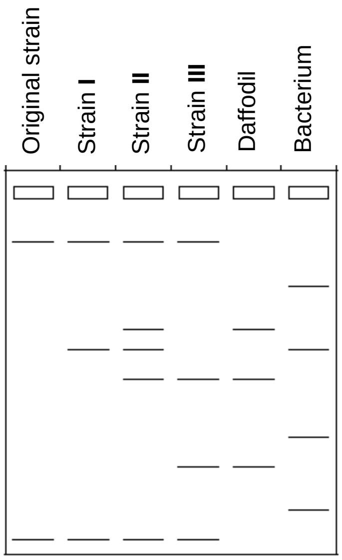
B The microorganisms may not survive.

C The mutation rate of the microorganisms would increase.

D The transfer of changed genes to other organisms.

30. An attempt was made to produce Golden rice. To determine whether or not DNA from the daffodils and the bacterium had been successfully incorporated in the DNA of the rice, scientists used PCR and gel electrophoresis to produce DNA profiles.

The following DNA profiles belong to the original strain of rice, three strains **I** to **III** of genetically modified Golden rice, and the species of daffodil and bacterium used to incorporate beta-carotene genes in the rice.



Which one of the strain(s) of Golden rice has successfully incorporated DNA from both the daffodil and the bacterium?

A Strain **I** only

B Strain **II** only

C Strain **I** and **III** only

D Strain **II** and **III** only

END OF PAPER